STORAGE AND BACK UP

Assignment 1: RAID concepts

RAID is a data-storage virtualization technology that combines multiple physical disks into logical units to increase performance, fault tolerance, or both.

RAID Levels:

- RAID 0 – Striping

  Function: When we write the data on a hard drive. The data gets split and gets written on both the hard drives.

  Benefits:

  High read/write performance

  Utilizes full disk capacity

  Drawbacks:

  No redundancy; failure of one disk = total data loss

  Use Cases: Temporary data, cache, non-critical workloads

- RAID 1 – Mirroring

  Function: The data we write gets copies into two disks.

  Benefits:

  High redundancy

  Quick recovery on disk failure

  Drawbacks:

  Storage overhead: 50% loss

  Use Cases: OS disks, databases requiring high availability.

- RAID 5 – Striping with Distributed Parity

  Function: Uses parity distributed across disks.

  Benefits:

  Fault tolerance for 1 disk

  Efficient storage utilization

  Drawbacks:

  Slow write performance

  Rebuild time is long

  Use Cases: File servers, read-heavy applications.

- RAID 6 – Striping with Double Parity

  Function: Two parity blocks on separate disks

  Benefits:

  Survives 2 simultaneous disk failures

  Drawbacks:

  Slower writes than RAID 5

  Use Cases: Large-capacity drives, archival storage.

- RAID 10 – Combination of RAID 1 + RAID 0

  Function: Striped mirrors

  Benefits:

  Best performance + redundancy

  Fastest rebuild

Drawbacks:

Expensive (50% usable capacity)

Use Cases: Databases, high-performance transactional apps.

Multipathing

Multipathing provides multiple physical paths between a server and storage.

Benefits

- High Availability: If one path fails, I/O continues on another.
- Load Balancing: Distributes I/O across multiple paths for better performance.
- Automatic Path Failover & Recovery
- Redundancy at Network, HBA, Switch, and Storage levels

Types

- Native Multipathing:
    - Eg: Linux DM-Multipath, Windows MPIO
- Vendor Multipathing:
    - Eg: EMC PowerPath, Hitachi HDLM, NetApp MPIO

Vendor tools often provide:

- Better monitoring
- Optimized algorithms
- Faster failover

Block Storage

- **How it works:** Data stored in blocks (raw volumes)
- **Protocols:** iSCSI, FC
- **Use Cases:**
    - Databases (Oracle, SQL, MySQL)
    - Virtual Machines (VMware, Hyper-V)
    - High performance transactional apps

File Storage

- **How it works:** Files organized in directories

- **Protocols:** SMB/CIFS, NFS
- **Use Cases:**
    - User home directories
    - Shared folders
    - Content repositories

Object Storage

- **How it works:** Data stored as objects with metadata
- **Protocols:** HTTP/S3
- **Use Cases:**
    - Cloud storage (AWS S3, Azure Blob)
    - Backups and archives
    - Media files, logs, big data

Benefits to Organizations

- Increased data reliability
- High performance
- Better scalability
- Reduced downtime
- Efficient data management
- Improved backup and disaster recovery readiness

Assignment 2: Brocade SAN

Switches

Switches connect servers, clients to storage in a Fibre Channel (FC) environment.

Types

- Brocade Switches (common in enterprise SAN)
- Cisco MDS Series Switches

Functions

- Manage FC traffic
- Provide zoning
- Enable multipathing

- Ensure secure and optimized storage connectivity

Brocade & Cisco – GUI/CLI

Brocade

- **GUI:**
  - Web Tools
  - Brocade Network Advisor (BNA) / SANnav
- **CLI:**
  - Used through SSH or console
  - Common commands:
    switchshow
    portshow
    zoneshow
    cfgshow

Cisco MDS

- **GUI:** Cisco Data Center Network Manager (DCNM)
- **CLI:**
  - Based on Cisco IOS
  - Common commands:
    show flogi database
    show interface fc1/1
    show zone

Zoning

Mapping of storage(LUNs) to the server is called zoning.

Types

- WWN Zoning (preferred): Uses HBA WWN
- Port Zoning: Based on switch port

Benefits

- Enhances security
- Reduces traffic interference
- Avoids accidental disk access
- ISOLATES Initiator–Target communications

Assignment 3: NetBackup and Rubrik Setup

NetBackup Overview & Architecture

Veritas NetBackup

An enterprise backup solution for VM, physical servers, databases, and cloud.

Architecture Components

- **Master Server:** Policy, catalog DB
- **Media Server:** Handles data movement
- **Client:** Servers being backed up
- **Storage:** Disk, Tape, Cloud

Rubrik Overview & Architecture

Rubrik

A cloud data management platform offering backup, restore, archival, and DR.

Architecture

- **Rubrik Cluster (Briks)** – nodes running backup services
- **Rubrik CDM** – management platform
- **APIs** for automation
- **Global file system** that deduplicates and compresses data

Backup Configuration (VM & Physical Servers)

For VMs (VMware/Hyper-V)

- Install backup agent (if required)
- Add vCenter/ESXi to backup console
- Create policies:
    - Schedule
    - Retention
    - Storage destination
- Use VMware snapshots for consistent backups

For Physical Servers

- Install backup client

- Select file system/application agents
- Configure:
    - Full backup
    - Incremental backup
    - Logs backup (for DB servers)

Backup Types

Full Backup

- Entire data every time
- Simple but time-consuming

Incremental Backup (INC)

- Backs up only data changed since last Incremental backup
- Fast, minimal storage usage

Differential Backup

- Backs up data changed since last Full backup
- Larger than incremental

Progressive Backup

- Only backs up changed files
- Combines incremental + archival concepts

Synthetic Full Backup

- Creates a full backup from previous full + incremental without touching production systems

NDMP Backup (Network Data Management Protocol)

- Used for NAS devices (NetApp, EMC Isilon, etc.)
- Data moves directly from **NAS → Tape/Backup storage**
- Reduces load on backup server